

Sparse diffusion models for multi-annotator medical image segmentation

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ABSTRACT

Medical image segmentation is crucial for accurate medical diagnosis and treatment planning, but it often suffers from ambiguity due to the inherent complexity of medical images and variability among annotators. Addressing this issue requires to capture and integrate the diverse insights from multiple annotators. Denoising diffusion models show advantages in predicting the annotations from multiple experts. However, the sampling process of these models is time-consuming. In this paper, we introduce a novel Sparse Diffusion-based Segmentation framework (SDSeg) for efficient multi-rater medical image segmentation, aiming to generate consensus masks at different levels of agreement among annotators. In the proposed framework, we build a Background-Adaptive Spatial Sparse (BASS) module that accelerates the inference process by concentrating computational efforts on critical regions within the image. The proposed module reduces the sampling time while maintaining high segmentation accuracy. Experimental results demonstrate the effectiveness of our method on five different datasets, showing significant improvements in segmentation reliability and efficiency compared to existing techniques. Code will be released via <https://github.com/lhaof/SparseDiffusion> after the acceptance.

1. Introduction

Precise segmentation of medical images enables the accurate identification of anatomical structures, including lesions [1], tumors [2], and organs [3,4], which is critical for achieving reliable medical outcomes. Typically, a group of experts or annotators independently delineate regions of interest within images, producing multiple segmentation results for a single image. The inherent complexity of medical images and the variability among different annotators often lead to uncertainty or ambiguity in the segmentation results [5]. The ambiguity limits the applicability of general segmentation network models and makes the algorithm development process more complex.

To address the challenge of multi-annotator prediction for medical image segmentation, common approaches include majority voting [6,7] or label fusion strategies [8]. These methods model the differences and agreements among various annotators by aggregating their individual labels, thus exploring the variability and reliability of multi-annotator segmentation. However, these approaches demonstrate limited segmentation accuracy. Alternative approaches aim to surpass the

limitations of deterministic segmentation models by incorporating uncertainty through various mechanisms, such as employing Bayesian neural networks [9] or Monte Carlo Dropout techniques [10], to produce multiple predictions and then averaging them. Nonetheless, these techniques may dropout key information and lack sufficient representation ability.

Recently, Denoising Diffusion Probabilistic Models (DDPM) [11] have shown exceptional performance in various computer vision tasks by simulating the process of incrementally adding noise to images and subsequently learning to reverse this process. Advancements in diffusion models have demonstrated their significant potential in image segmentation, especially within the realm of medical imaging [12–16]. Some recent studies [17–19] have validated the effectiveness of diffusion models for multi-annotator medical image segmentation. However, the segmentation process of these models is inefficient and time-consuming because generating a single output requires multiple forward passes.

We observe that the regions of interest in medical images often occupy only a small area of the complete image. The feature

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computation for most background regions can be redundant during the inference process. To reduce the sampling time for individual samples, we develop a Sparse Diffusion-based Segmentation (SDSeg) framework for handling multi-annotator medical image segmentation. The SDSeg framework integrates two sub-networks: Prior Net and Denoising Net. The Prior Net generates soft masks supervised by the average of annotations from multiple raters, and introduces features into the other sub-network. The Denoising Net produces consensus masks at various levels through a series of iterative denoising steps. A consensus mask at level c is a segmentation mask whose foreground pixels are agreed by at least c annotators. In the Denoising Net, we design a Background-Adaptive Spatially Sparse (BASS) module that eliminates the need to process most background or non-active image patches. During the inference stage, this BASS module can directly replace the decoder layers of the Denoising Net without retraining, thereby enabling faster segmentation.

In sum, our work has three main contributions:

- We design an efficient Sparse Diffusion-based Segmentation framework based on the sparse regions of interest (ROIs) in medical images, accelerating the segmentation process while keeping high performance.
- We propose a Background-Adaptive Spatially-Sparse module that primarily processes foreground-region features and accelerates the proposed segmentation framework during the inference stage.
- We validate the effectiveness of our proposed method on five multi-annotator MRI (Magnetic Resonance Imaging) and CT (Computed Tomography) datasets across four different anatomical regions. The proposed framework achieves state-of-the-art segmentation results with better efficiency.

2. Related work

2.1. Multi-annotator medical image segmentation

Deep learning models for automated medical image segmentation have achieved remarkable success [20–25]. Early work [26] typically used deterministic models with a single fused mask as ground truth, failing to fully leverage the uncertainty inherent in multiple annotations. To address this, Bayesian neural networks [9,27] and Dropout techniques [10] were introduced to model uncertainty by learning distributions of network weights or by introducing randomness during inference. However, these methods often result in pixel-level variability and unstable outputs. To generate more diverse and realistic predictions, conditional variational autoencoders have been combined with U-Net architectures [28–30], but these approaches still lack sufficient diversity or require complex network structures [30].

Recently, generative diffusion models have provided a new perspective on handling ambiguity by simulating the data generation process through Markov chains. Existing methods based on diffusion models [17–19,31] have shown improved capabilities in capturing intra- and inter-rater variations, generating multiple plausible segmentation results. We consider that multi-rater labels can be represented as multiple consensus masks which high-light foreground regions agreed by at least a specific number of annotators. Thus, our work is to predict multiple segmentation maps at different consensus levels for an input image.

2.2. Denoising diffusion model

Denoising diffusion models have shown exceptional capabilities across various domains, notably in tasks such as image synthesis [11, 32–34], restoration [35], editing [36–38] and feature learning [39]. These models leverage Markov chains to progressively transform data into noise, which is then reversed through a denoising process to generate high-fidelity images or other types of data. The advanced models are producing more realistic outputs with enhanced stability and

diversity [40,41]. Diffusion models have been applied to image segmentation [42,43]. Some methods [12,44] predict class labels from features extracted by diffusion models, while some others [14,45] directly employ conditional diffusion models for binary segmentation. Additionally, some other works [46,47] have focused on refining segmentation masks. Another recent work [48] employs sparse masked modeling to learn general features, while our approach focuses on accelerating the feature extraction process using sparse convolutions. In this paper, we adopt the diffusion process to model multi-rater labels, and exploit the sparsity of foreground regions for efficient segmentation. The proposed technique reduces computational costs while maintaining the model size and capacity, achieving efficient and accurate image segmentation.

3. Methodology

3.1. Problem definition

The task of multi-annotator medical image segmentation involves N expert annotators independently identifying and labeling various structures or pathological patterns within the same image $I \in \mathbb{R}^{H \times W \times C}$, where H , W , and C represent the height, width, and number of channels of the image, respectively. For each image, the labeling process yields a series of diverse segmentation annotations $\{M_i\}_{i=1}^N$, reflecting the varying understandings and interpretations of the image content by different N experts.

3.2. Network architecture

To effectively incorporate the agreement and disagreement information among multiple raters, we propose a new Sparse Diffusion-based Segmentation (SDSeg) framework for multi-annotator medical image segmentation. Fig. 1 illustrates the overall architecture of our method, which contains two distinct sub-networks, Prior Net and Denoising Net. An input image I is first fed into the Prior Net, which generates a soft mask $\mathbf{m}_p$, providing an initial probabilistic estimate of the segmentation. The Prior Net utilizes a U-shaped convolutional architecture that employs a five-level encoder-decoder structure with skip connections. The soft mask is supervised using the average segmentation labels from different annotators. The loss function is given as:

$$\mathcal{L}_p(\mathbf{m}_p, \mathbf{M}) = \mathbb{E}[\|\bar{\mathbf{m}} - \mathbf{m}_p\|^2], \quad (1)$$

where $\mathbb{E}$ represents the expectation operator and $\|\cdot\|^2$ denotes the mean squared error. $\bar{\mathbf{m}}$ denotes the average segmentation label and can be computed as $\frac{1}{N} \sum_{i=1}^N M_i$.

The Denoising Net is a UNet with skip connections based on some previous work [19], and has seven layers in both its encoder and decoder. Each layer in the Denoising Net contains 3 normal residual blocks. Multi-head attention layers are adopted to connect two adjacent residual blocks in the last encoder layer and the first decoder layer of the Denoising Net. The embedding of time step is sent into each residual block as condition. The Denoising Net starts from a noisy map and progressively denoises it until obtaining the segmentation map. Due to the uncertainty in medical image segmentation caused by inter-rater variability, we use consensus levels of multiple annotators as a condition to generate the corresponding segmentation map [19]. For a segmentation map at c consensus levels, each of its foreground pixels is labeled by at least c annotators. The foreground regions of a low-consensus mask exactly contain those regions of higher-consensus masks. Given the segmentation labels of N experts, we could compute the ground truth consensus mask at N levels. Let $\mathbf{m}^c$ denote the ground-truth (GT) consensus mask at level c . $\mathbf{m}^c$ can be calculated as:

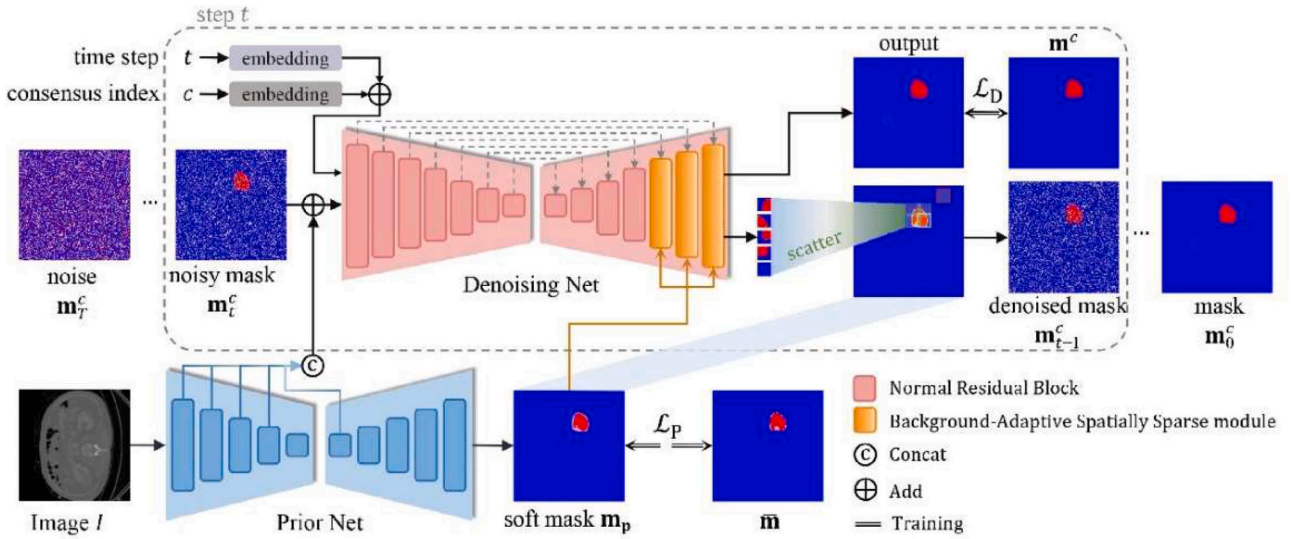


Fig. 1. Illustration of our proposed Sparse Diffusion Segmentation (SDSeg) framework. The framework consists of a Prior Net and a Denoising Net. The Prior Net produces image features for the Denoising Net to generate segmentation mask. For training, the framework uses normal residual blocks. During inference, the proposed Background-Adaptive Spatially-Sparse (BASS) modules take the soft mask m_p as the input to identify the RoI and reduce the redundant computations.

$$m^c(x, y) = \begin{cases} 1, & \sum_{i=1}^N M_i(x, y) \geq c, \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

where x and y denote the coordinates of each pixel in the consensus mask. To introduce the diffusion process, we consider a time step $t \in [1, T]$ and a consensus index $c \in \{1, 2, \dots, N\}$. With the paired data $\{\text{Image } I, \text{Mask } m_0^c\}$, we train a conditional diffusion model [32] to estimate the denoised result m_c corresponding to the consensus index c .

The diffusion model uses a T-step forward process to add noise to the ground truth of a consensus map, and then adopts a T-step reverse process to denoise the noisy map. The forward process adds t -step normal noise ϵ to the clean consensus map m_0^c , producing the noisy output m_t^c as:

$$m_t^c = \sqrt{\alpha_t} m_0^c + \sqrt{1 - \alpha_t} \epsilon, \quad \epsilon \sim N(0, I), \quad (3)$$

where $\alpha_t := 1 - \beta_t$, $\bar{\alpha}_t := \prod_{s=1}^t \alpha_s$, $\{\beta_t\}_{t=1}^T$ is the noise schedule and ϵ is sampled from the normal distribution.

We first process the noisy consensus map m_t^c with a 2D convolution layer and then add it to the features from the Prior Net, shown as $\oplus$ in Fig. 1. The features from the Prior Net are generated by upsampling the encoder features from 4 scales and concatenating them along the channel dimension, displayed as $\odot$ in Fig. 1. The time step index t and consensus index c are embedded and then integrated before passing through each residual block in the Denoising Net. The Denoising Net is required to predict a map without noise, and is trained with the diffusion loss $\mathcal{L}_D = E_{m_t^c, t, \epsilon} [\|m_0^c - D(m_t^c, t, c)\|_2^2]$, where D denotes the Denoising Net. The overall loss function is defined as the sum of the diffusion loss and the prior loss: $\mathcal{L} = \mathcal{L}_D + \mathcal{L}_P$.

In the reverse process, the Denoising Net iteratively denoises a noise map starting from step T , until we obtain a clean consensus map which is conditioned by a consensus index. At each time step t , we use Eq. (4) to compute m_{t-1}^c :

$$m_{t-1}^c = \frac{1}{\sqrt{\alpha_t}} \left(m_t^c - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} D(m_t^c, t) \right) + \sigma_t z, \quad (4)$$

where $z \sim N(0, I)$. For inferring an image, the Prior Net only forwards once to provide image features and a soft mask. For each consensus index, we predict e consensus masks by repeatedly running the model,

and these e masks are averaged to yield a consensus mask at the given index. And then we take average of the N consensus masks at N different levels to produce a final soft-label map.

3.3. Background-adaptive spatially-sparse module

For existing diffusion models, their iterative sampling scheme presents a significant challenge due to its high computational cost. To address the issue, we introduce the Background Adaptive Spatially Sparse (BASS) module to reduce running time. As illustrated in Fig. 2, BASS accelerates the sampling process by applying convolution operations only to specific regions of the features, thereby reducing unnecessary computations. The core is to leverage the inherent spatial redundancy found in many images, particularly in background areas that typically contain less critical information for the generation process. BASS has three components: Active Patch Sampling (APS), Sparse Residual Block (SRB), and Background-Adaptive Normalization (BAN). These components work together to effectively reduce time while maintaining high-quality segmentation results. The following provides a detailed description of each component.

3.3.1. Active patch sampling

APS is to identify and select the critical regions of feature maps based on the soft mask, output by the Prior Net. Considering the relatively high quality of the output soft mask m_p , we identify the minimum bounding box that encloses all foreground areas. Then, we find the closest ratio (r_h, r_w) of the bounding box of the foreground region from the predefined aspect ratio set $\mathcal{R} = \{(1, 2), (2, 1), (1, 3), (3, 1)\}$. The bounding box is then divided into $n_h \times n_w$ image patches. n_h and n_w are determined by:

$$(n_h, n_w) = \gamma \cdot (r_h, r_w), \quad \gamma = \max\{\text{round}(2h/H), \text{round}(2w/W), 1\}, \quad (5)$$

where h/w and H/W denote the height/width of the bounding box and the entire image, respectively. γ is a factor controlling the number of patches. For each sampled patch, it is discarded if it contains only background pixels. The left-top coordinates of each patch are referred as *Active Indices* that means the corresponding patch is active during the inference. All the sampled patches are represented as a set of active indices:

$$S = \{(x_1, y_1), (x_2, y_2), \dots, (x_k, y_k)\}, \quad (6)$$

where $k = n_h \times n_w$. By utilizing the coordinates in S , we can crop the

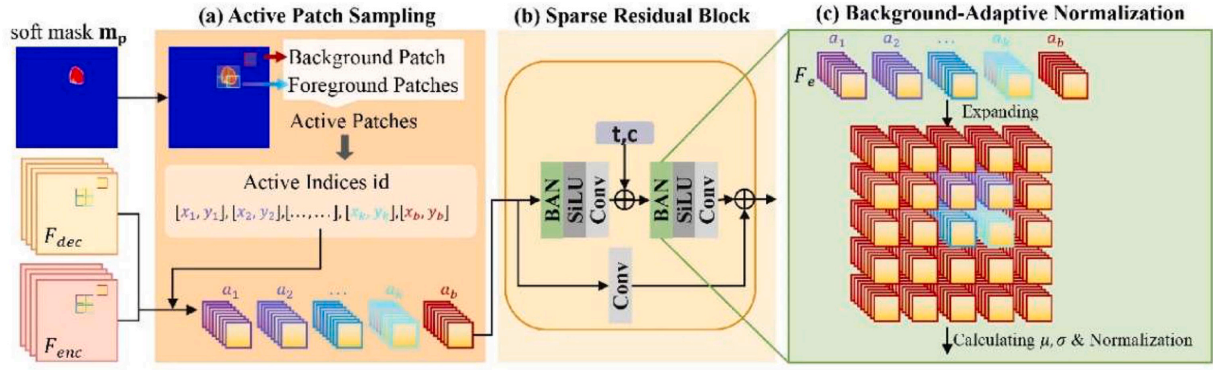


Fig. 2. Overview of the Background-Adaptive Spatially-Sparse (BASS) module. (a) Active Patch Sampling is to extract position indices of active patches from a soft mask, and sample patch-size feature maps from input features F_{enc} and F_{dec} . (b) Sparse Residual Block is to apply convolution to sampled patch features and condition embeddings. (c) Background-Adaptive Normalization is to approximate the normalization in full-size feature map for processing the sparse patch features.

corresponding feature regions at the specified locations and then apply convolutions and other operations to these cropped feature maps.

3.3.2. Sparse residual block

Following the sampling of patch-level feature maps, the BASS module integrates a Sparse Residual Block (SRB) to facilitate efficient feature learning. The SRB takes a background-patch and multiple foreground-patch features as input, and outputs enhanced and normalized patch features. Specifically, the SRB consists of three convolution layers. A Convolution layer estimates a residual from the input, which is subsequently added to the output to strengthen feature representation. The other two convolution layers are sequentially stacked, with each layer preceded by a Background-Adaptive Normalization (BAN) operator and a SiLU activation function. Additionally, the embedding features of t and c are incorporated into the patch features between these two convolution layers.

During the training phase, we do not use the BASS module; instead, we update the Denoising Net based on normal residual blocks. In the inference phase, we replace the normal residual blocks at the last three scales in the decoder of the Denoising Net with the proposed BASS modules, and copy the parameters from the replaced residual blocks into the BASS modules. As a result, the proposed framework with the BASS module can directly perform efficient segmentation without further fine-tuning or retraining.

After computing the final layer in the Denoising Net, we crop the foreground patches of the output according to the active indices and fill these patches in the soft mask m_p to obtain the segmentation result. To ensure seamless integration between the foreground outputs and the surrounding background, padding is applied to patch features during the sampling process, and the padded margins are trimmed before filling in the soft mask.

3.3.3. Background-adaptive normalization

Generative diffusion models commonly employ Group Normalization [49,32,50] which dynamically computes the mean and variance across separate channel groups for each sample. These normalization layers learn parameters and statistical calculations from full-size feature maps, which is inappropriate for our proposed BASS module using selective patch-level features. The discrepancy between the statistics computed on full-size feature maps during training and the patch features during inference may lead to a cumulative bias, particularly affecting the feature computation for each foreground region.

To address the above issue, we propose the Background-Adaptive Normalization (BAN) technique. Considering that the background remains relatively homogeneous within a time step, we replicate a single background patch feature for u times to approximate the entire background region. This replication allows us to calculate a more accurate

mean and variance, where the combined area of the u replicated patches is approximately equal to the area of the entire background region. These replicated background-patch indices (x_b, y_b) are incorporated into the set (defined in Eq. (6)) to obtain the set S that contains $(k + u)$ pairs of indices. The parameter u represents the number of background patches contributing to the normalization process. These u replicated background patches are normalized together with the foreground patches, ensuring that the feature values output by the BASS modules more accurately reflect the actual distribution of the features being processed by normal residual/convolution blocks.

The input feature is denoted as $F_e \in \mathbb{R}^{(k+1, C, p, q)}$, where $k + 1$ is the number of active patches, C is the number of channels, and p and q are the height and width of active patches, respectively. We expand the background patch features by a factor of u to obtain $F' \in \mathbb{R}^{(k+u, C, p, q)}$, and then compute the mean μ_{group} and variance σ_{group}^2 for each channel group. The feature values belonging to the same channel group share the same mean and variance. The Normalization process can then be expressed as:

$$\text{BAN}(F_e) = w \frac{F_e - \mu_{group}}{\sqrt{\sigma_{group}^2 + \varepsilon}} + b, \quad (7)$$

where ε is a stability constant. w and b are learnable scaling and shifting parameters, respectively.

By adaptively normalizing with a background patch, BAN ensures that the computed statistics better reflect the actual distribution of the features being processed. The proposed normalization mitigates the bias introduced by original group normalization, ensuring more accurate and consistent feature representations of sparse regions.

4. Experiments

4.1. Dataset

The proposed framework is evaluated using the data from the Quantification of Uncertainties in Biomedical Image Quantification (QUBIQ) challenge [51]². The challenge includes five datasets of MRI and CT images across four different anatomical regions. Each dataset focuses on a binary segmentation task related to specific pathologies or healthy anatomical structures, and is annotated by various experts to provide professional segmentation results. These datasets include the Kidney Segmentation dataset (CT), the Brain Growth Segmentation dataset (MRI), the Brain Tumor Segmentation dataset (multi-modal MRI), the Prostate Segmentation dataset (MRI), and the Liver Tumor Segmentation dataset (CT). To split the dataset into training and testing sets, we follow the official division provided by the challenge. For the Kidney Segmentation dataset, the ratio of training and testing samples is 5:1. For the other datasets, we set the ratio as 7:1 following the official

setting of the challenge.

4.2. Implementation details

We implement the proposed framework and conduct experiments on a Tesla V100 GPU with 32 GB memory. The framework is trained using the AdamW optimizer with a learning rate of $2e-5$, and all computations are performed in float32 precision. During the training, gradients from the Denoising Net are blocked from propagating back to the Prior Net. The selective features from the Prior Network are determined based on the validation set, a small subset separated from the training set. The hyper-parameters of diffusion model are set as ACP [19], including the number of diffusion steps $T = 100$, the number of ensemble $e = 25$, and the data augmentation strategy. For all experiments, we use standard DDPM sampling and set the image size to 256×256 pixels. Considering that large-size features bring in high computational cost and that deep decoder features contain low-level information, we applied our background-adaptive spatially sparse modules at the decoder features of the resolutions [64, 128, 256], as shown in Fig. 1.

We adopt the same benchmarks and evaluation metric (soft Dice) as previous multi-annotation prediction methods [19]. We use multiple threshold levels (0.1, 0.3, 0.5, 0.7, 0.9) for the soft Dice. The Dice metric was calculated at each threshold level on the binarized ground truth and predictions. The final performance score was obtained by averaging the Dice scores across all thresholds. This comprehensive evaluation method ensures a robust assessment of our framework's segmentation capabilities.

4.3. Results and discussion

4.3.1. Comparison with the state-of-the-art methods

To validate the reliability of segmentation results on multi-annotator datasets, we compared our proposed method with other various existing medical image segmentation methods on five datasets from QUBIQ. Table 1 includes a range of methods such as convolutional neural networks (e.g., FCN, MCD), models employing multi-rater strategies such as majority-vote U-Net (MV-UNet), label sampling (LS-UNet), multi-head strategies (MH-UNet), and the state-of-the-art general segmentation model Rolling-UNet, which was adapted by training with majority-vote ground truth. Additionally, our method is compared with the models designed for multi-rater segmentation tasks (MRNet and TAB), and with diffusion-based methods (AMIS and CCDDM) that randomly select one from multiple labels for training. We evaluated methods like DMISE, MedSegDiff, HiDiff, and ACP, which are similar to our approach and based on consensus masks.

Table 1

Comparison among existing medical image segmentation methods and ours on QUBIQ datasets. D_K , D_{BG} , D_{BT} , D_{P1} and D_{P2} represent the Kidney, Brain Growth, Brain Tumor, Prostate.1 and Prostate.2 datasets, respectively. The best Soft Dice for each dataset is in bold and the second is underlined.

Method	D_K	D_{BG}	D_{BT}	D_{P1}	D_{P2}
FCN [52]	70.03	80.99	83.12	84.55	67.81
MCD [10]	72.93	82.91	86.17	86.40	70.95
MV-UNet [20]	70.65	81.77	84.03	85.18	68.39
LS-UNet [53]	72.31	82.79	85.85	86.23	69.05
MH-UNet [54]	73.44	83.54	86.74	87.03	75.61
Rolling-UNet [55]	93.72	91.97	92.93	92.60	83.58
MRNet [56]	95.12	89.55	89.73	93.43	83.61
TAB [57]	93.95	91.95	90.96	93.50	86.88
AMIS [17]	68.53	74.09	92.95	91.64	21.91
CCDDM [18]	96.24	92.53	92.49	87.67	73.46
DMISE [58]	93.97	92.89	93.38	93.47	85.15
MedSegDiff [15]	95.36	91.55	92.12	92.70	83.06
HiDiff [59]	94.51	71.01	91.51	93.72	85.96
ACP [19]	<u>96.70</u>	91.95	<u>93.94</u>	<u>94.96</u>	<u>88.52</u>
Ours	97.02	<u>92.64</u>	94.06	95.14	90.40

As shown in Table 1, our method obtained the highest Dice scores on four datasets, and the second best results on the other one dataset. Diffusion-based segmentation methods achieved notable results in multi-rater medical segmentation tasks. Particularly in prostate segmentation, our method achieved the highest Dice scores on Prostate.1 and Prostate.2, outperforming the specialized multi-rater strategy method TAB by 1.64 % and 3.52 %, respectively. The above results indicate that our approach not only surpasses traditional and multi-rater strategies but also outperforms most diffusion models, especially in handling annotator inconsistencies. Fig. 3 shows the visual comparison between existing approaches and ours. The proposed method produces less noises out of the segmented objects, it may be due to that the BASS module helps focus on foreground regions.

To study the sampling speed of diffusion models, we show the average time required to infer an image and the segmentation performance of different methods. As illustrated in Fig. 4, non-diffusion models (Rolling-UNet, MRNet, TAB) can rapidly generate segmentation results but often at the cost of lower Dice scores. In contrast, our method not only provides robust predictions, but also reduces the inference time by 35 % to 57 % compared to our baseline without using sparsity ('w/o sparse' in Fig. 4).

We further report the running time and Dice scores of competitive diffusion-based methods in Table 2. As Table 2 shows, using the BASS almost does not affect the sDice (see a small drop of 0.5 % in Prostate.2 dataset and a slight increase of 0.19 % in Brain Growth dataset.). Notably, using the BASS module can significantly reduce the computing time by 35 % - 57 % for one sample.

Compared to three existing diffusion-based methods in Table 2, our proposed method with BASS modules obtains the highest sDice scores on four of the five datasets, and shows the shortest running time on all datasets. Besides, our method demonstrates remarkable computational efficiency in terms of TFlops per sample. Specifically, our approach with BASS requires only an average of 15 TFlops across different datasets, which is less than 10 % of the TFlops of DMISE, MedSegDiff and ACP. When comparing our method with and without BASS, the integration of BASS further reduces the computational complexity from 28 TFlops to as low as 15 TFlops per sample, achieving a reduction of about 50 % while maintaining comparable segmentation accuracy. The efficiency improvement is due to that our method adaptively focuses the computation on regions of interest. In contrast, existing diffusion-based segmentation methods perform inference on the feature of an entire image, resulting in redundant computations in background regions and consequently higher costs. Interestingly, the enhance of efficiency does not lead to any noticeable drops in the quality of segmentation, and even brings in slight improvements in segmentation performance. Overall, our method effectively addresses annotator inconsistency and efficiently generates high-quality segmentation results.

4.3.2. Ablation study

In Table 3, we validated the strength of the internal mechanisms in our method (denoted as M5). The results from Prior Net (M1) are inferior to those of M2 and M5, highlighting the effectiveness of the Denoising Net and sparse strategy in our work, respectively. Comparing M2 to M3 with a sparse strategy reveals a significant performance drop, particularly the ~ 25 % decrease on the kidney dataset, when sparsity is introduced without utilizing background information. M4 improves the performance of M3 by using background-guided normalization that adopts a single background patch without expansion. The improvement of M4 shows the importance of background data in the feature normalization. Applying Background Adaptive Normalization (BAN) allows M5 to closely match M2, demonstrating that BAN effectively introduces sparsity to reduce inference time while maintaining accuracy. As the comparison between M2 and M5 in Table 3 shows, using the sparse mechanism with the BAN layer (namely, the BASS module) does not affect the segmentation results much (such as slightly reducing Dice by 0.5 % or increasing Dice by 0.33 %).

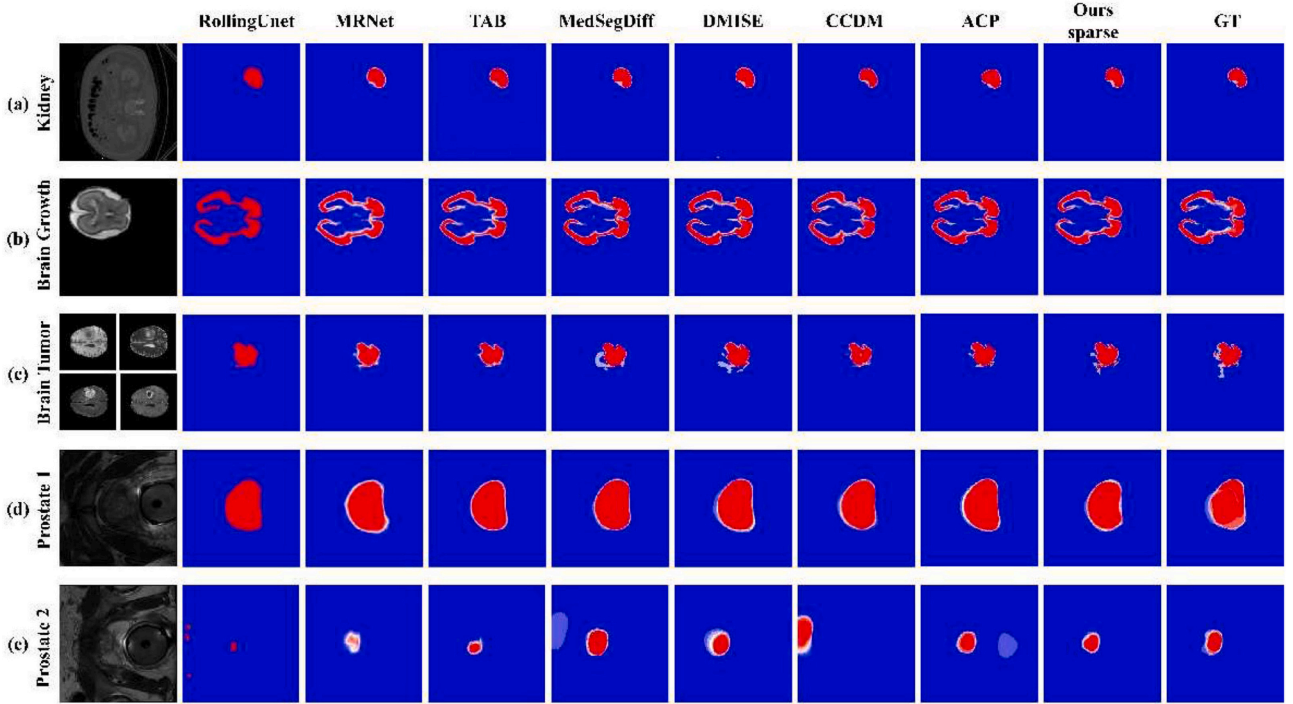


Fig. 3. Visual comparison among different methods on QUBIQ datasets. Each row shows a testing case from a dataset and the leftmost column presents the input image. The rightmost column is ground truth (GT), and the other columns display the segmentation results from various methods. Segmentation masks are visualized using the 'bwr' colormap with blue ('background') and red ('foreground').

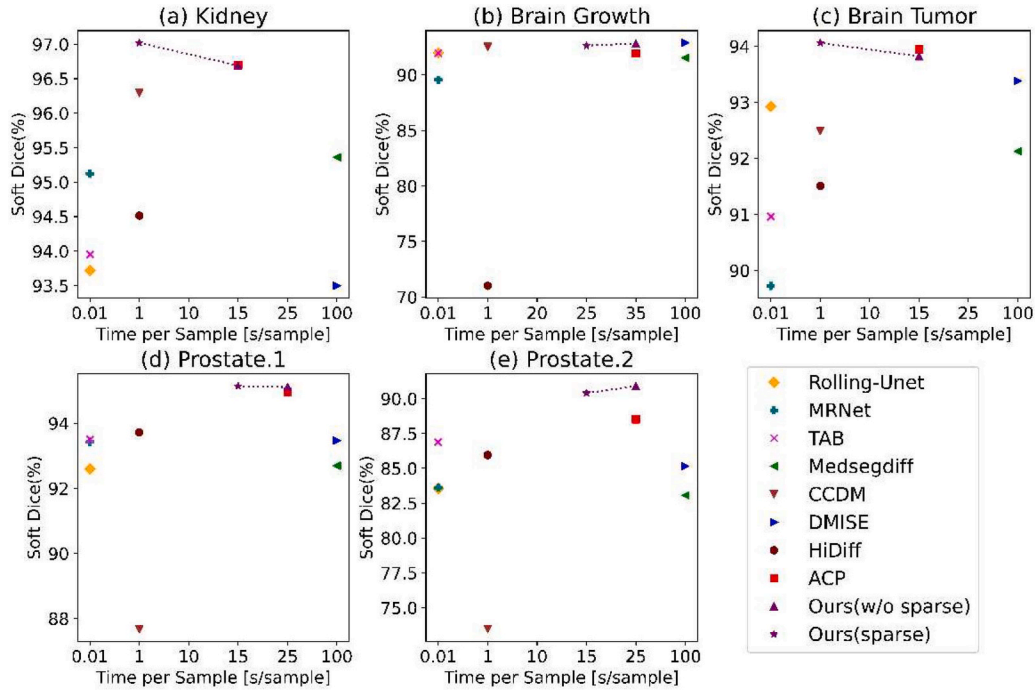


Fig. 4. Segmentation accuracy and efficiency of previous methods and ours on QUBIQ datasets. The x-axis represents the average inference time per sample (in seconds) using GPU, while the y-axis indicates the Soft Dice score. Each marker corresponds to an image segmentation method. Dashed lines connect our proposed methods using and without using the BASS.

4.3.3. Effect of the number of segmentation times per sample

The number of segmentation times per sample (denoted as e) is a crucial hyper-parameter for generating reliable segmentation results, and is studied in this subsection. During the inference, each testing sample with a noise image is input into the model to produce a

segmentation output. To improve the stability and reliability of the segmentation, for the same testing sample we run our proposed model for e times via feeding e different noise images and obtain multiple distinct segmentation results. These e results are then combined to generate the final segmentation result, via pixel-wise averaging and

Table 2.

Efficiency and accuracy of diffusion-based medical image segmentation methods. D_K , D_{BG} , D_{BT} , D_{P1} and D_{P2} represent the Kidney, Brain Growth, Brain Tumor, Prostate.1 and Prostate.2 datasets, respectively. The highest sDice, the lowest time and TFlops are in bold. Avg. means taking average of TFlops of all datasets, since the TFlops of most methods are constant.

Dataset	Metric	Method				
		DMISE	Medsegdiff	ACP	Ours (w/o BASS)	Ours
D_K	sDice	93.97	95.36	96.70	96.69	97.02
	Time	152.65	176.07	19.51	18.91	8.18
D_{BG}	sDice	92.89	91.55	91.95	92.83	92.64
	Time	323.75	372.88	41.37	39.63	25.85
D_{BT}	sDice	93.38	92.12	93.94	93.82	94.06
	Time	166.25	169.91	19.52	18.87	8.92
D_{P1}	sDice	93.47	92.70	94.96	95.11	95.14
	Time	305.90	304.08	35.90	34.30	17.93
D_{P2}	sDice	85.15	83.06	88.52	90.90	90.40
	Time	284.27	315.66	35.53	34.43	16.96
Avg.	TFlops	496	260	169	28	15

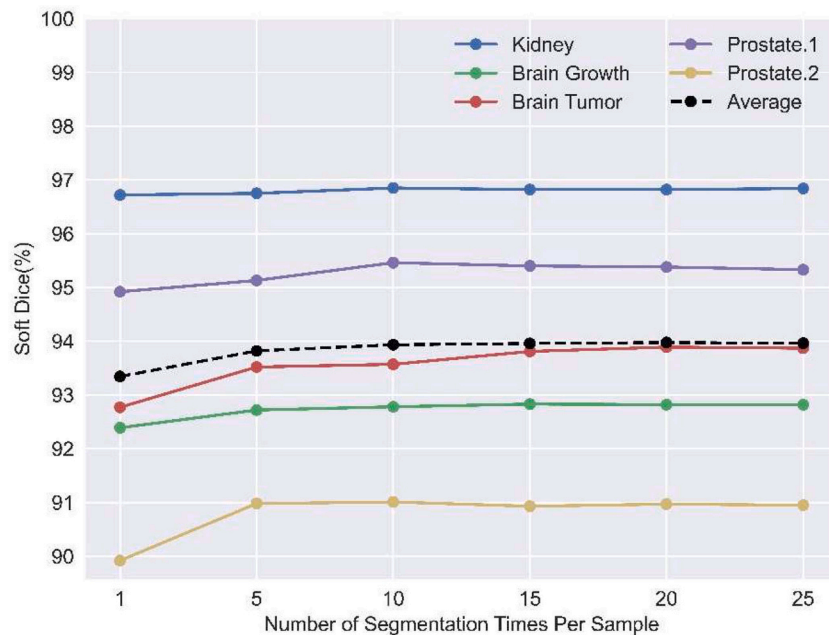
rounding. For each noise image, the proposed method predicts c masks of different consensus levels. These c consensus masks are averaged after aggregating the e results.

The above-mentioned technique can be viewed as an ensemble of results obtained from different input noise maps, allowing the model to leverage variations in predictions to achieve stable performance. As illustrated in Fig. 5, the results show a clear pattern across all datasets.

Table 3

Ablation study of our proposed method. M1 shows the results from Prior Net. M2 with DM denotes our baseline model with a Denoising Net using normal residual blocks, and ‘DM’ means using Diffusion Models. ‘Sparse’ means using the BASS module normalizing foreground features without background sampling. BGN refers to Background Guided Normalization using one background-patch feature for normalizing, without expanding. BAN is our proposed Background Adaptive Normalization.

	DM	Sparse	BGN	BAN	D_K	D_{BG}	D_{BT}	D_{P1}	D_{P2}
M1					93.44	89.49	91.68	90.58	82.72
M2	✓				96.69	92.83	93.82	95.11	90.90
M3	✓	✓			71.59	92.54	81.35	87.20	83.06
M4	✓	✓	✓		77.15	92.88	87.86	90.67	85.98
M5	✓	✓		✓	97.02	92.64	94.06	95.14	90.40

**Fig. 5.** The effect of different numbers of segmentation times per sample on Soft Dice for our method.

The performance improves as the number of ensemble samples increases, and then gradually stabilizes. Additionally, the fact that satisfactory results are achieved even with fewer ensemble iterations highlights the efficiency of our approach.

4.3.4. How annotator inconsistency affect the segmentation performance?

To provide quantitative evidence for the robustness of our method in handling annotator inconsistencies, we performed further analysis. We first quantified the inter-annotator agreement for each test sample using the minimum pairwise Dice score among any pairs of available ground-truth masks, and a lower value indicates lower annotator consistency. We then plotted this consistency metric against the segmentation performance for our method and other diffusion-based methods. As shown in Fig. 6, the results clearly demonstrate that while the performance of all methods degrades with lower annotator agreement, our method consistently maintains a superior performance margin across the entire range of label quality. These results suggest that SDSeg is not only highly accurate but also more robust to the label noise and variability inherent in real-world multi-annotator datasets.

4.3.5. How ROI-to-image ratio affect the efficiency?

To validate the design of the BASS module, we investigated the source of its efficiency gains. We analyzed how the acceleration effect scales with the sparsity of the input data by plotting the inference time reduction (%) against the ground-truth ROI-to-image ratio for each sample in the test set. As shown in Fig. 7, the speed-up is most significant for small ROIs (reducing up to 60 % inference time) and diminishes as

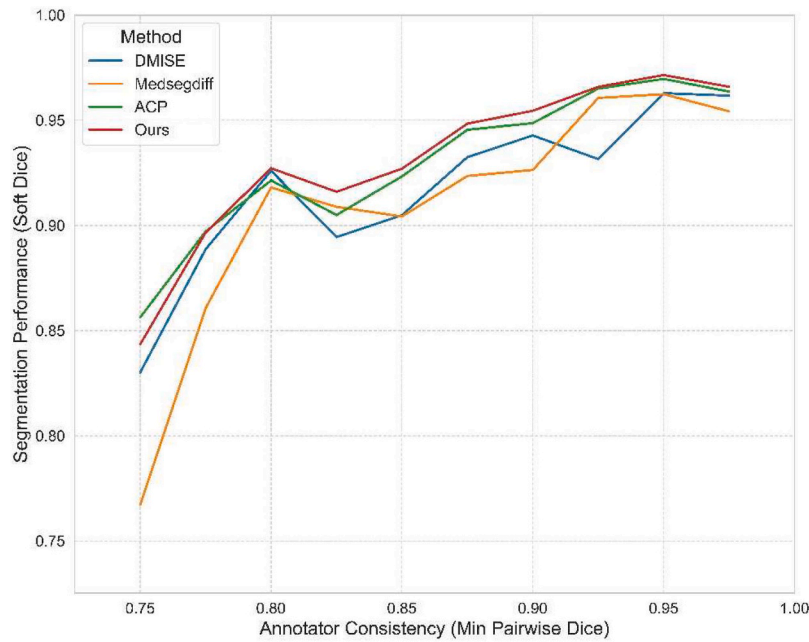


Fig. 6. Segmentation performance with annotator consistency. The x-axis represents inter-annotator consistency, measured by the minimum pairwise Dice score among all annotations for a given sample. The y-axis is the segmentation performance. The trend lines illustrate the average performance trend for each method. Our method consistently outperforms other methods.

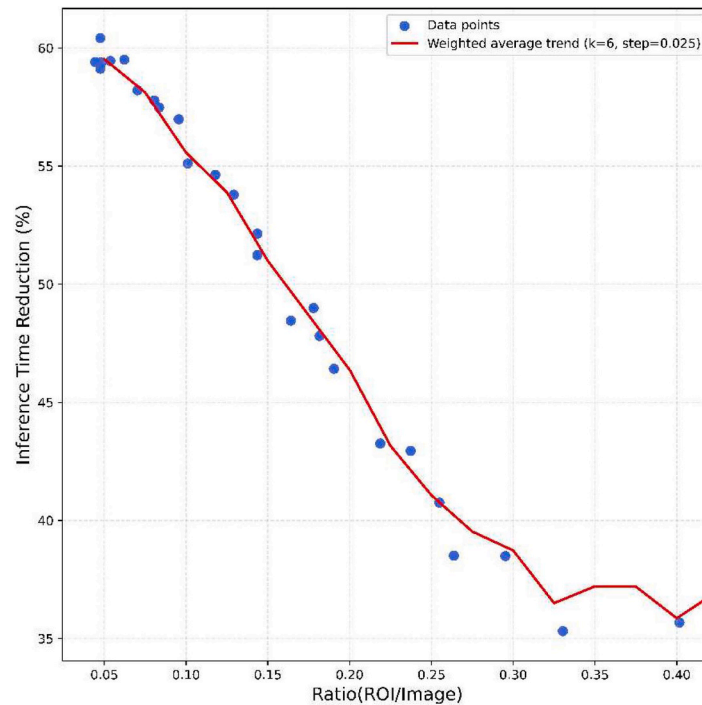


Fig. 7. Inference time reduction vs. the ROI-to-image ratio. This plot demonstrates that the acceleration from the BASS module is greatest for smaller ROIs and decreases as the ROI size increases. The trend line represents the average trend of the data points.

the ratio increases. Notably, as the ROI-to-image ratio becomes larger than 30 %, our proposed BASS module can still reduce at least 35 % inference time and show stable computation efficiency with increasing ROI size. The above analysis confirms that the efficiency of the BASS module is based on the inherent sparsity of the medical imaging task.

4.3.6. Limitations and failure cases

In Fig. 8, we have included several typical failure cases which could

cause the proposed BASS module to produce degraded results. These samples can be categorized into two groups. First, in Row 1 & Row 3 of Fig. 8, the contrast between the foreground lesion and the background region is so low that our Prior Net could predict incomplete or shrunk foreground area which is not well refined by our Denoising Net. Second, in Row 2 of Fig. 8, when the signals or patterns of foreground target are unnoticeable, like only one of six annotators labeling the segmentation region, it is also difficult for our Prior Net to locate the initial foreground

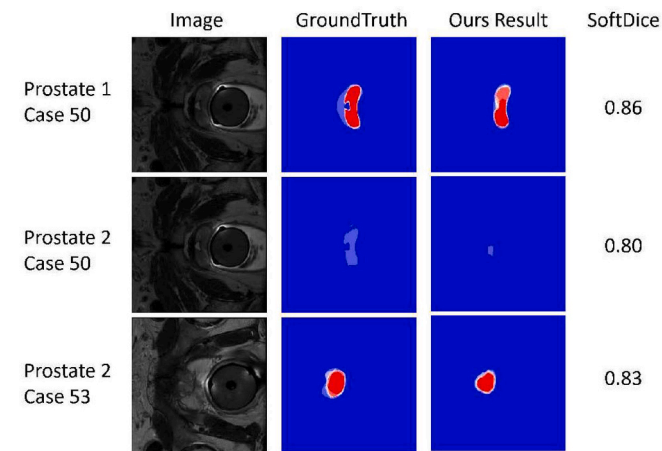


Fig. 8. Qualitative results of challenging cases from the test sets. The proposed SDSeg show relatively lower performance in these cases. Such cases often involve low lesion-to-background contrast, which can challenge the initial Prior Net prediction and subsequently affect the final segmentation quality. The SoftDice score for each case is provided on the right.

region accurately. Future work could address such failures by improving the information provided by the Prior Net and enhancing the refinement capability of the segment diffusion model.

Another limitation of our proposed method is that our approach is only designed for binary-class multi-annotator segmentation task. In future works, our proposed method could be adapted to multi-class applications. For example, the multi-class segmentation task can be regarded as multi-channel binary segmentation. Active patches can be sampled from multiple prior-mask channels corresponding to multiple categories, and the background region can be defined as the reverse of union of multi-channel active patches. In such a way, the proposed SDSeg framework with the BASS module may adapt to multi-class segmentation tasks.

Besides, since the proposed SDSeg framework is based on the sparsity of ROIs, including the sparse number of foreground pixels and objects, the proposed method has limitations in segmenting dense objects. For example, most histopathology image segmentation tasks aim to segment densely distributed cell nuclei or tumor regions containing many disconnected parts. These dense foreground regions require to process many active foreground patches and make it difficult to select a representative background patch. The complex background may also pose challenges to the approximation of background features.

5. Conclusion

In this paper, we introduced an efficient Sparse Diffusion-based Segmentation (SDSeg) framework for multi-annotator medical image segmentation, addressing the redundant feature computation in background regions of input images. The proposed framework integrates a Prior Net to provide an initial estimate of segmentation, and a Denoising Net to produce multiple segmentation masks at different consensus levels. We also developed a novel Background-Adaptive Spatial Sparse (BASS) module for efficient inference, which reduces sampling time while maintaining high performance. Experimental results have shown the effectiveness of the Denoising Net and the proposed BASS module in our SDSeg model.

Future work could explore integrating the proposed spatially-sparse module into the training phase of the diffusion model, which may help ensure the consistency between training and inference stages to further enhance the segmentation performance. Besides, in future we will try to evaluate our method on more image modalities (like ultrasound or endoscopy) when their multi-annotator benchmarks are available, and try to collect high-quality multi-rater annotations of more image

modalities for evaluation.

CRedit authorship contribution statement

Yang Ji: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Haofeng Li:** Writing – review & editing, Project administration, Methodology, Data curation, Conceptualization. **Guanbin Li:** Writing – review & editing, Supervision, Formal analysis. **Siqi Liu:** Writing – review & editing, Formal analysis, Data curation. **Xiang Wan:** Writing – review & editing, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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